

Table Notes

Road Accident Analysis SQL Database

Database Name

road_accident

Purpose of the SQL Database

The SQL database was created to support structured analysis of the Road Accident dataset. The cleaned CSV file was first imported into a staging table and then divided into three related tables. This structure helps demonstrate how the dataset can be organized relationally and analyzed using SQL queries, joins, filters, and aggregations.

The main relationship key used across the final tables is:

accident_index

1. Table Name: road_accident_stage

Purpose

This is the temporary staging table used to import the complete cleaned CSV file into MySQL before splitting it into final relational tables.

Key Column

accident_index

Important Columns

- accident_index
- accident_date
- day_of_week
- junction_control
- junction_detail
- accident_severity
- latitude
- longitude
- number_of_casualties
- number_of_vehicles
- speed_limit
- weather_conditions

- road_surface_conditions
- road_type
- vehicle_type

Note

Most columns in the staging table were imported as VARCHAR to avoid CSV import errors related to date, time, and numeric formatting. The correct data types were applied later while inserting data into the final tables.

2. Table Name: accident_fact

Purpose

This is the central fact table of the database. It stores the main accident-level facts such as accident date, time, severity, casualties, vehicles involved, speed limit, and calculated helper fields.

Primary Key

accident_index

Important Columns

Column Name	Description
accident_index	Unique accident record ID
accident_date	Date on which the accident occurred
accident_time	Time of accident
day_of_week	Day of the week when the accident happened
accident_severity	Severity of accident: Slight, Serious, Fatal
number_of_casualties	Number of casualties in the accident
number_of_vehicles	Number of vehicles involved
speed_limit	Speed limit of the road
year	Year extracted from accident date
month	Month name extracted from accident date
month_number	Month number used for sorting months
hour	Hour extracted from accident time
severity_score	Numeric severity score

Column Name	Description
serious_fatal_flag	1 if accident is Serious/Fatal, otherwise 0

3. Table Name: accident_location

Purpose

This table stores location-related accident details such as latitude, longitude, district, police force, and whether the accident happened in an urban or rural area.

Primary Key

accident_index

Foreign Key

accident_index references accident_fact(accident_index)

Important Columns

Column Name	Description
accident_index	Unique accident record ID
latitude	Latitude coordinate of accident location
longitude	Longitude coordinate of accident location
local_authority_district	Local authority district where accident occurred
police_force	Police force area related to the accident record
urban_or_rural_area	Indicates whether the accident occurred in an urban or rural area

4. Table Name: accident_conditions

Purpose

This table stores road, weather, light, junction, hazard, road surface, and vehicle-related details for each accident.

Primary Key

accident_index

Foreign Key

accident_index references accident_fact(accident_index)

Important Columns

Column Name	Description
accident_index	Unique accident record ID
junction_control	Type of junction control
junction_detail	Details of the junction where accident occurred
light_conditions	Lighting condition during the accident
carriageway_hazards	Hazards present on the carriageway
road_surface_conditions	Road surface condition during the accident
road_type	Type of road where the accident happened
weather_conditions	Weather condition during the accident
vehicle_type	Type of vehicle involved in the accident

Relationship Structure

The final database contains three main analysis tables connected by accident_index.

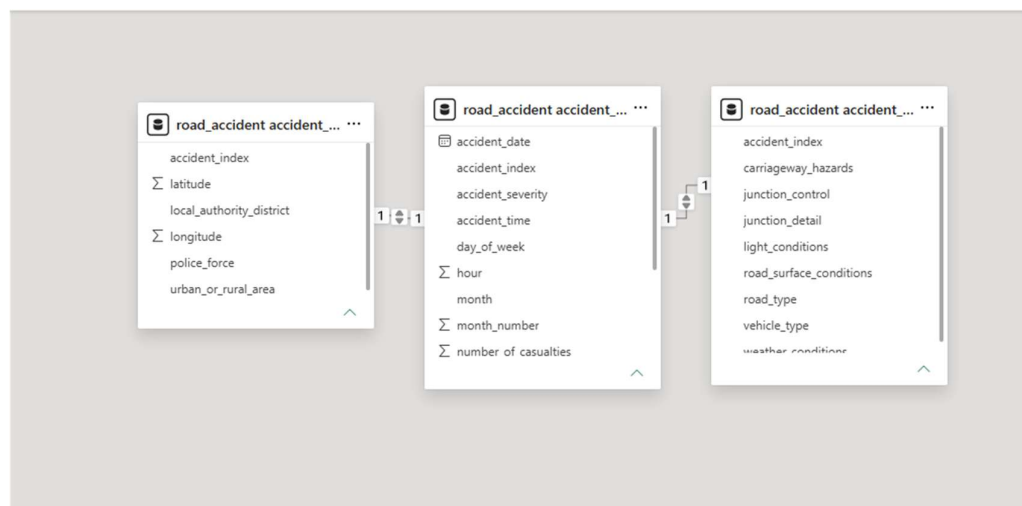
Relationship 1

accident_fact.accident_index = accident_location.accident_index

Relationship 2

accident_fact.accident_index = accident_conditions.accident_index

Simple Relationship Diagram



Explanation of Relationships

accident_fact is the central table because it contains the main accident-level information such as accident severity, casualty count, number of vehicles, speed limit, date, month, and year.

accident_location is connected to accident_fact using accident_index, allowing location-based analysis such as accidents by district, police force, and urban/rural area.

accident_conditions is also connected to accident_fact using accident_index, allowing analysis of road type, weather condition, light condition, road surface condition, and vehicle type.

This structure allows SQL JOIN queries to combine accident severity, casualty count, location, road type, weather, light condition, and vehicle type for business analysis.

Example JOIN Logic

```
SELECT
    f.accident_index,
    f.accident_date,
    f.accident_severity,
    f.number_of_casualties,
    l.local_authority_district,
    l.urban_or_rural_area,
    c.road_type,
    c.weather_conditions,
    c.vehicle_type
FROM accident_fact f
JOIN accident_location l
    ON f.accident_index = l.accident_index
JOIN accident_conditions c
    ON f.accident_index = c.accident_index;
```

Summary

The Road Accident SQL database was designed using one staging table and three final related tables. The structure makes the dataset easier to analyze and supports SQL queries using SELECT, WHERE, JOIN, GROUP BY, COUNT, SUM, and AVG.

This schema supports the main business objective of identifying accident severity patterns, casualty risk, road condition impact, vehicle involvement, and location-based accident trends.